

COMPUTER SCIENCE



Computer Organization and Architecture

Introduction of COA

Lecture_02



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**TOPICS
TO BE
COVERED**

o1

Instruction Cycle

o2

System Bus

Computer Generation

CA & CO.

Component of the Computer

① CPU

② Memory

③ I/O.

- ✓ PC
- ✓ AR / MAR
- ✓ DR / MBR / MDR
- ✓ IR
- ✓ AC
- ✓ SP
- ✓ PSW
- ✓ GPR.

Instruction Cycle.

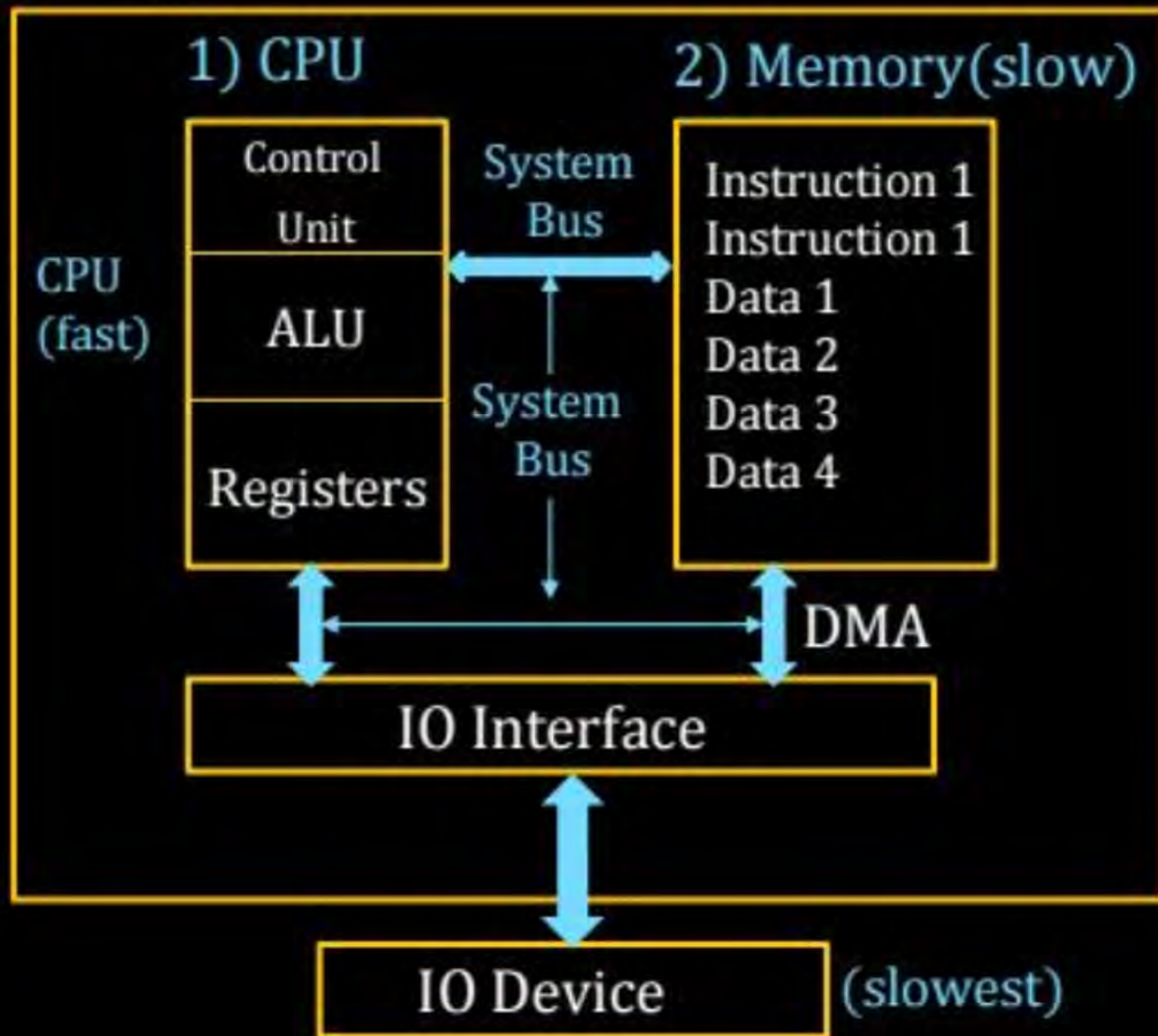
The Process Required for each Instruction Execution is called Instruction Cycle.

- ① Fetch Cycle
- ② Execute Cycle.

Component of Computer



- (1) CPU
- (2) Memory
- (3) Input/output





Instruction Cycle

Instruction Cycle



- ① Fetch cycle
- ② Execute cycle

Instruction Cycle



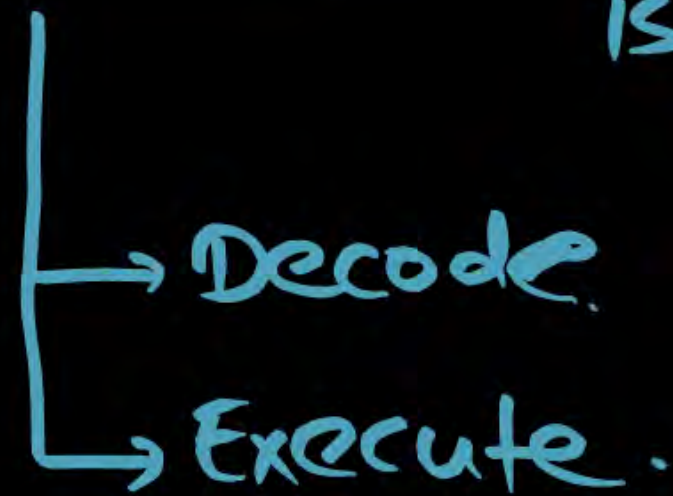
① Fetch cycle : To Fetch (bring) the Instruction from Memory to CPU (IR).

At the end of Fetch Cycle, PC is updated (Increment),
Now PC will Denote the Next Instruction Starting address.

Instruction Cycle



② Execute Cycle : The objective of Execute cycle is to process the Fetch Instruction.



Instruction Cycle

The process required to execute the Instruction.

(or)

Instruction cycle describe the execution sequence of the instruction.

Instruction Cycle contain 2 sub cycle.

- 1) Fetch cycle
- 2) Execute cycle
 - Decode
 - Execute

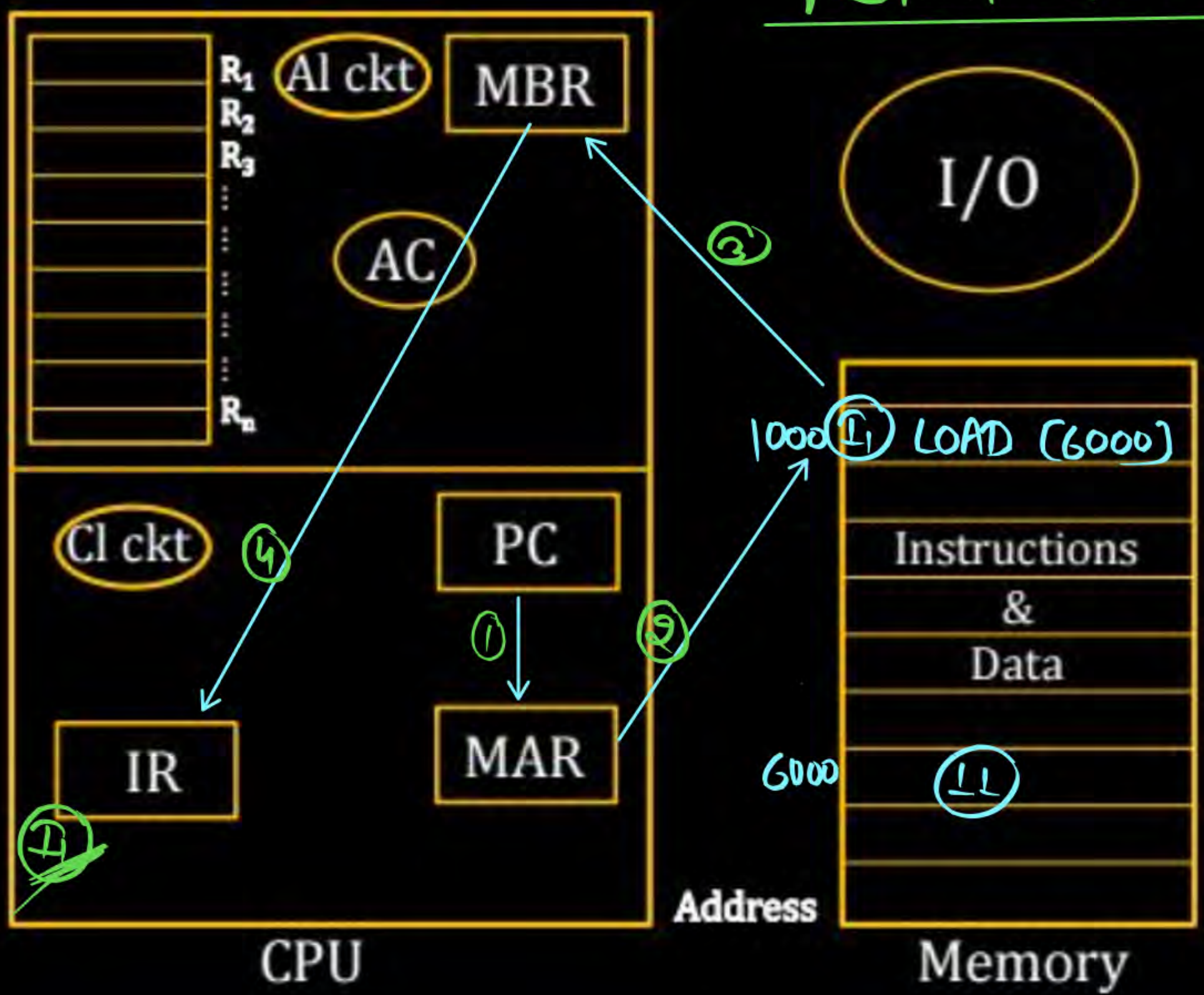
Instruction Cycle



Steps in Fetch cycle.



Fetch cycle.

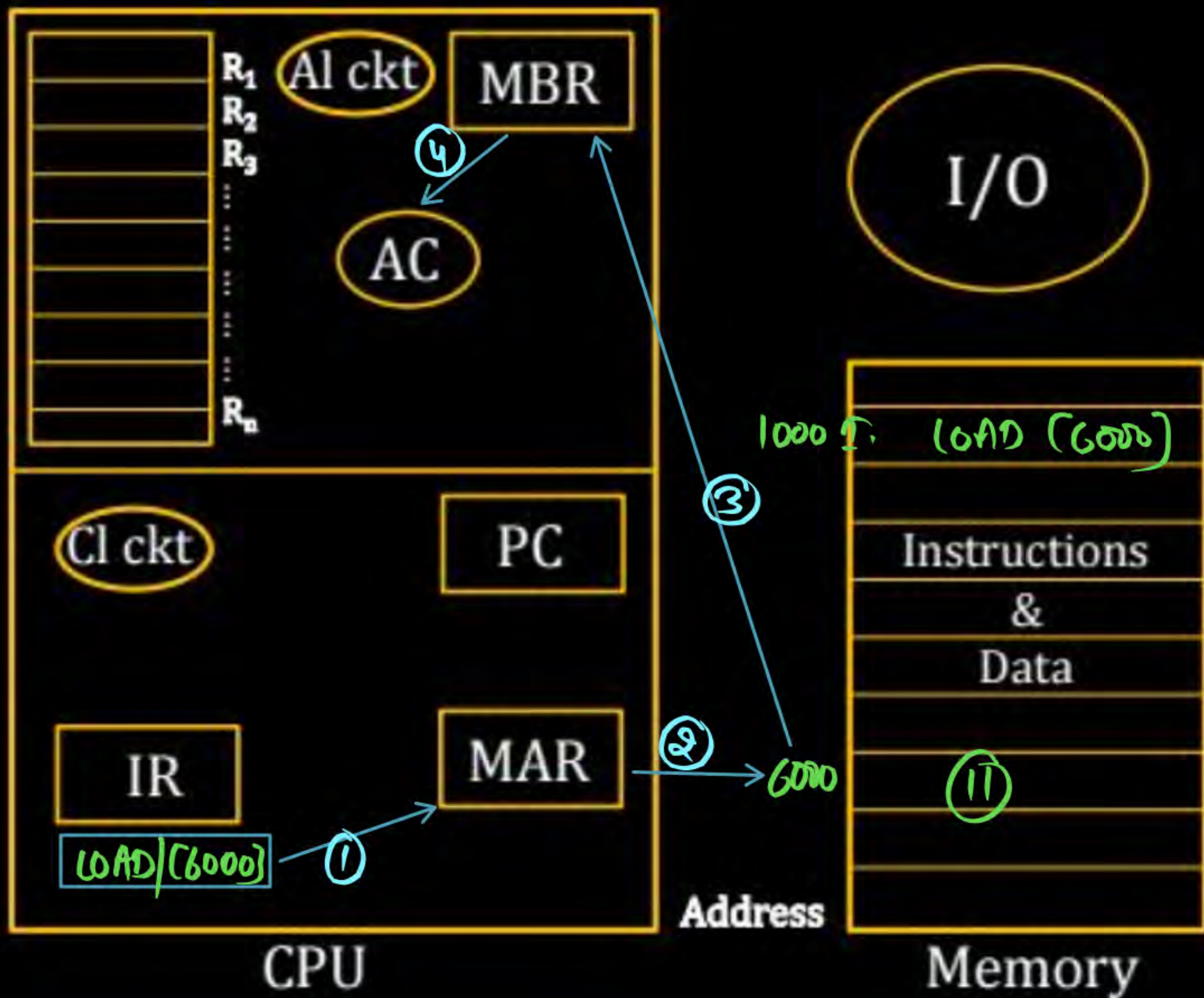


MEM to CPU (IR)

1000 I₁: LOAD (6000)

$AC \leftarrow M[6000]$

[LD] LOAD: memory Read.
[ST] STORE: memory Write.



1000: I, LOAD (6000)
 $AC \leftarrow M[6000]$

Steps in Instruction Cycle



Fetch
cycle

- 1) IAC (Instruction Address Calculation)
- 2) IF (Instruction Fetch)

[MEM to CPU (IR)]

Decode

- 3) Decoding (Analysis of Instruction
(What opcode, how many operand,
where operands are available))
- 4) OAC (Operand Address Calculation)
- 5) OF (Operand Fetch)

Execute
cycle

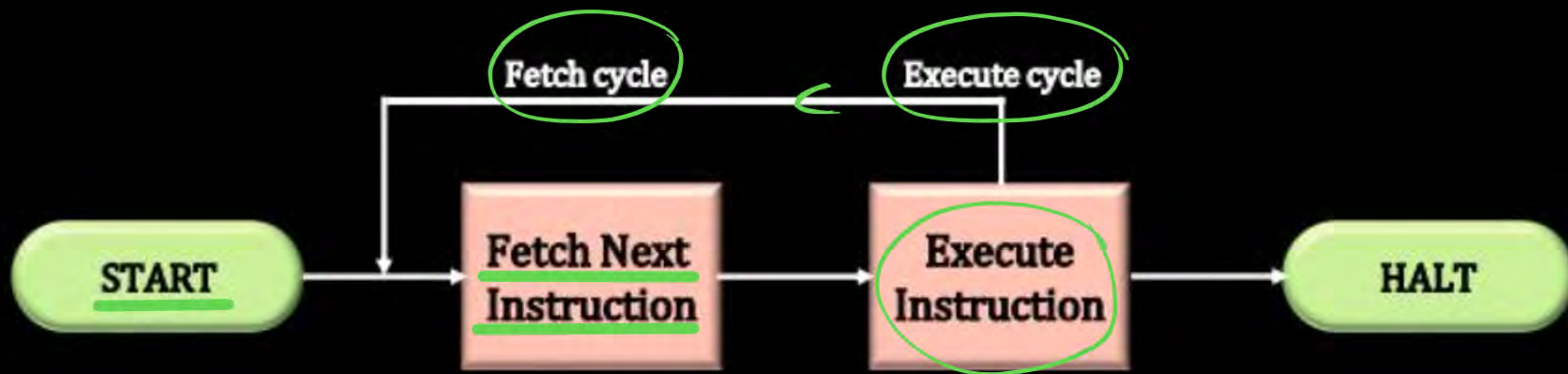
Execute

- 6) DP (Data Processing)
- 7) Result Storage

Fetch Cycle



- ❑ At the beginning of each instruction cycle the processor fetches an instruction from memory
- ❑ The program counter (PC) holds the address of the instruction to be fetched next
- Note* ❑ The processor increments the PC after each instruction fetch so that it will fetch the next instruction in sequence
- ❑ The fetched instruction is loaded into the instruction register (IR)
- ❑ The processor interprets the instruction and performs the required action.



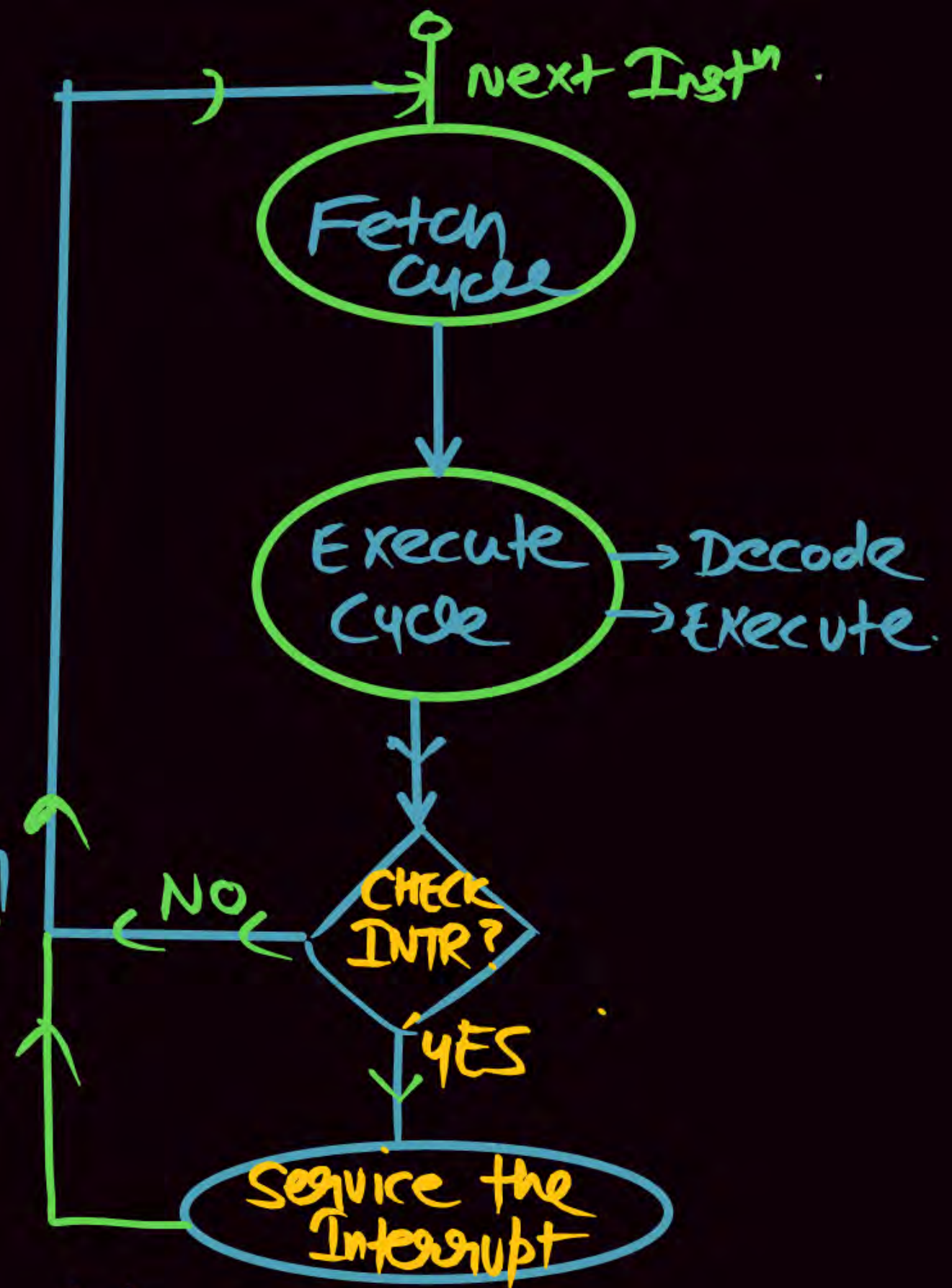
BASIC INSTRUCTION CYCLE

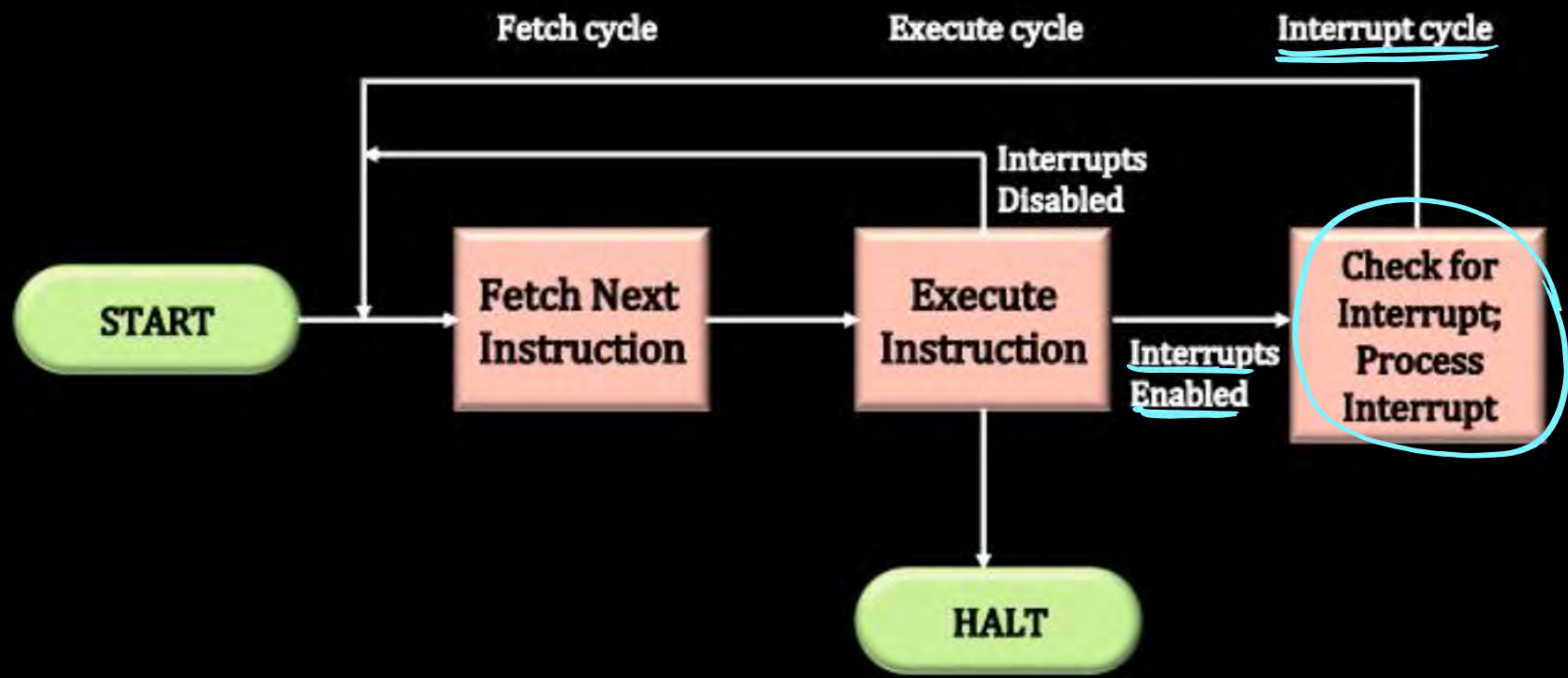
Instruction Cycle With Interrupt

- ① Fetch Cycle
- ② Execute Cycle
- ③ Interrupt Cycle

Note

AFTER Completion of the Instruction
Interrupt Will be Serviced.





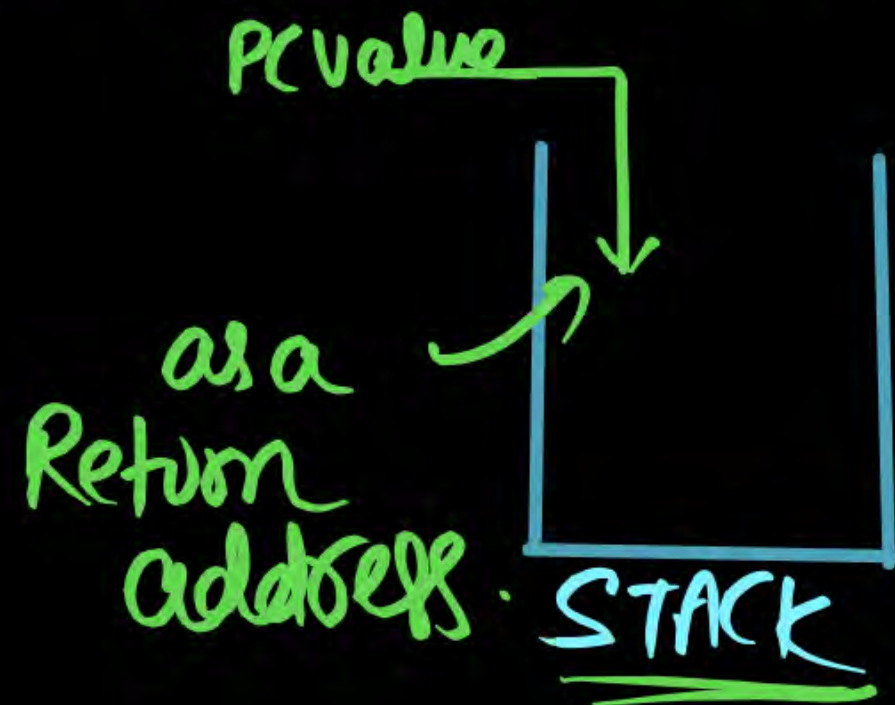
Instruction cycle with Interrupts



When Interrupt occur. then After Completion of Current Instruction execution. interrupt will be serviced.

When Interrupt occur. it PUSH the PC Value into the Stack as a Return address & Controlled transferred to ISR

(Interrupt Sub Routine)



& Controlled transferred to ISR.

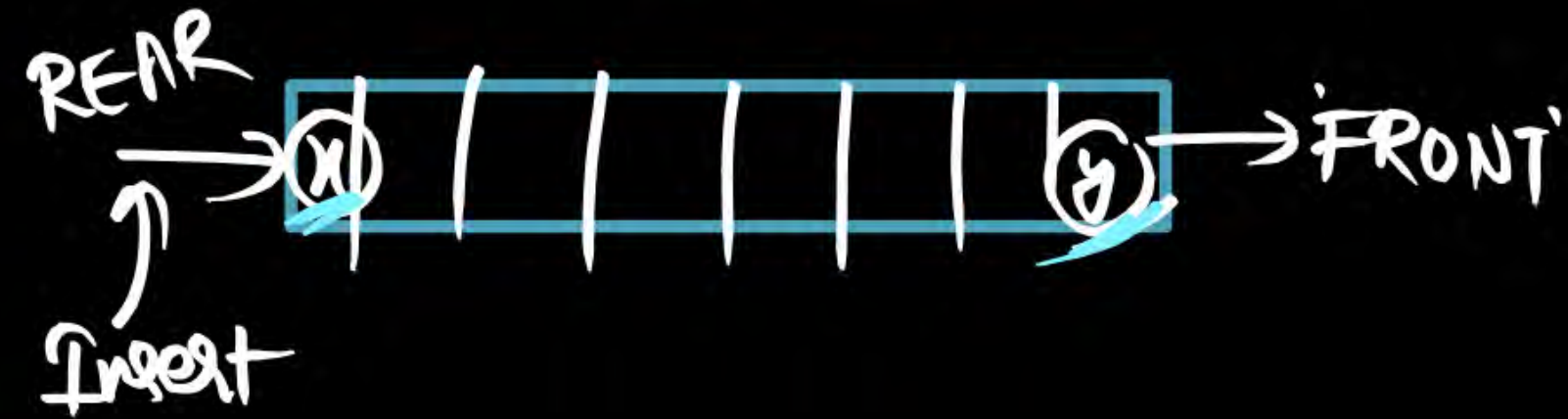
Q WHY IN STACK ?

Soln LIFO [Last in First out]



Why Not in Queue.

FIFO



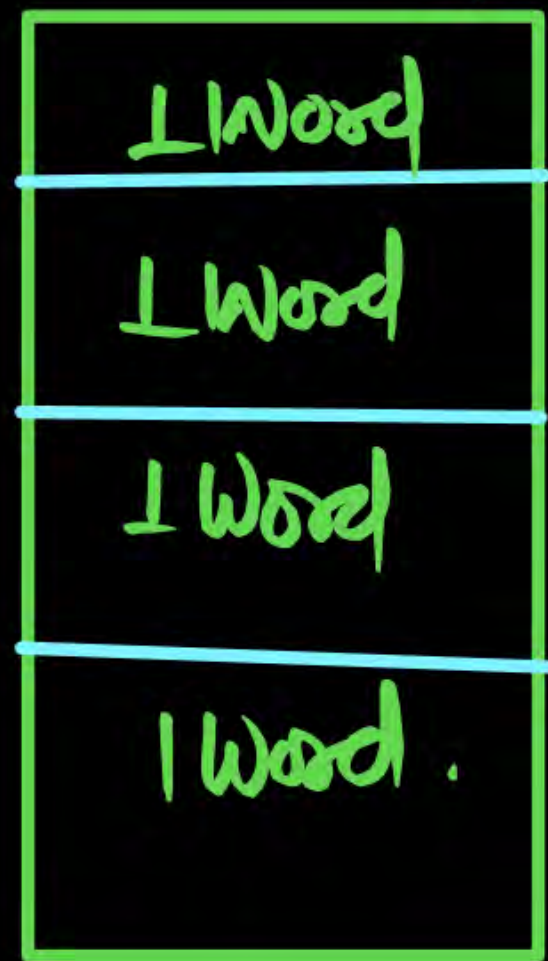
MEMORY

① Byte Addressable memory



1 cell = 1 Byte (8bit)

② Word Addressable memory



1 cell = 1 Word

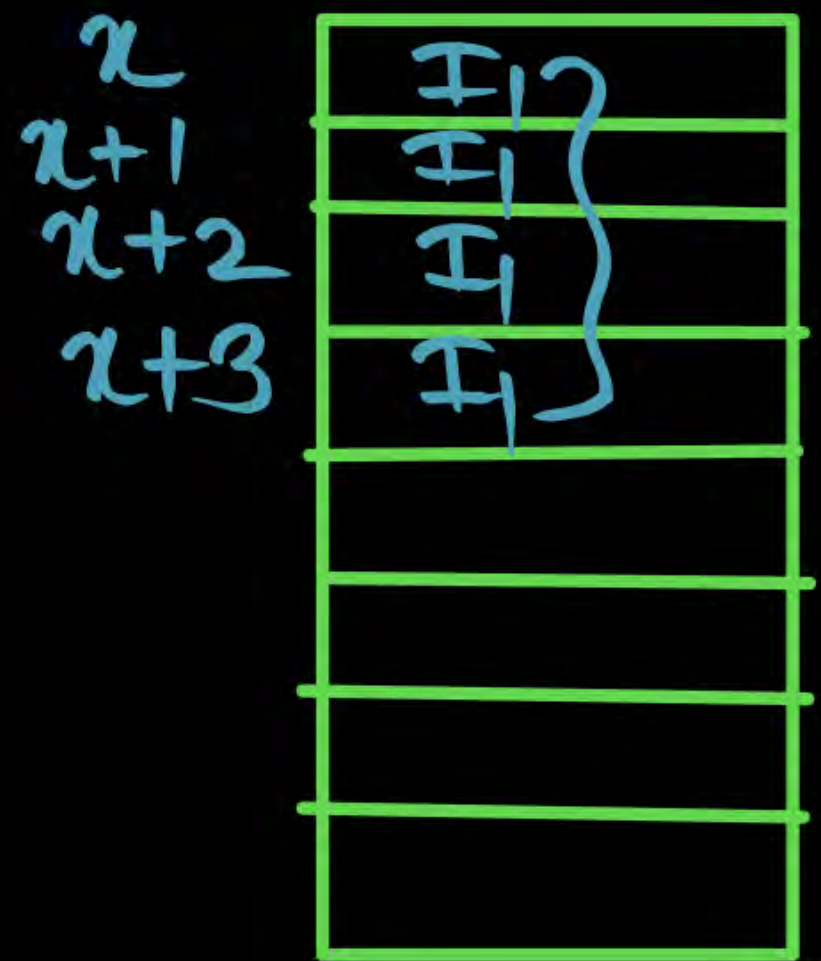
1 Word Size = 32 bit

1 Word = 4 Byte

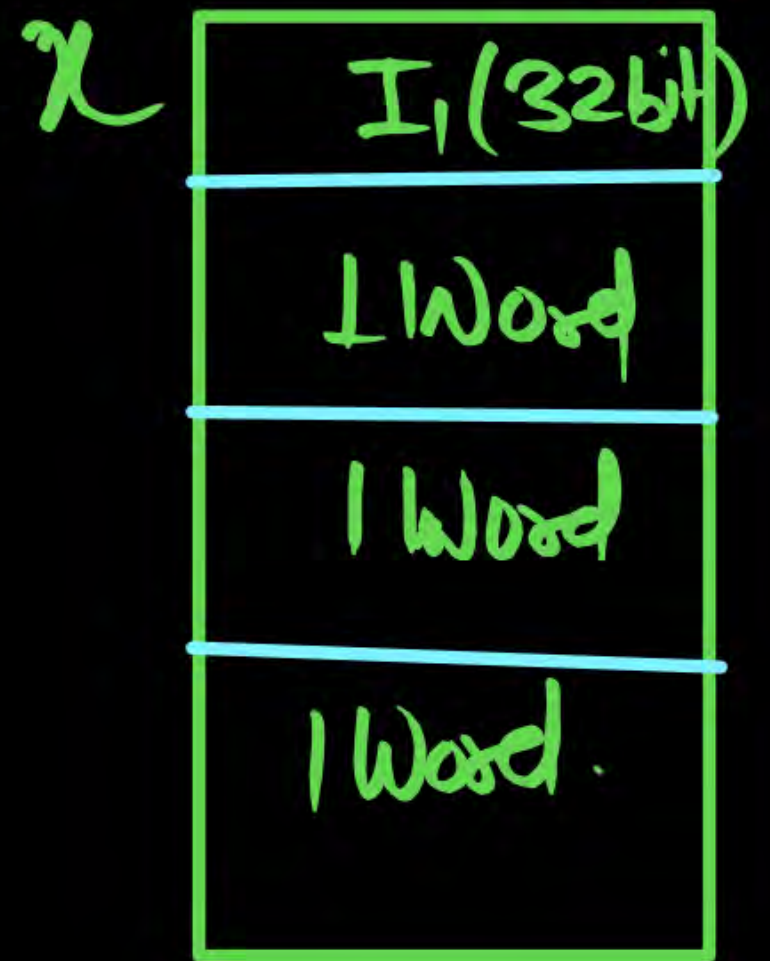
I_1 : 1 Word $\equiv$ 4 Byte

MEMORY

① Byte Addressable memory



② Word Addressable memory.





1 Kilometer = 1000 meter.

1 Word = x bit



$$5.6 \text{ km} \Rightarrow \underline{5600 \text{ meter}}$$

$\downarrow$

$$5.6 \times 1000 \Rightarrow$$

$$5600 \text{ meter} \Rightarrow \frac{5600}{1000} = \underline{5.6 \text{ km}}$$

es

1 Word = 32 bit $\approx$ 4 Byte

I₁: 2 Word

✓ I₁: 2 Word
 $\Rightarrow 2 \times 4 \text{ Byte}$
= 8 Byte

8 Byte Word?
 $\frac{8 \text{ Byte}}{4 \text{ Byte}} \text{ Words} \Rightarrow \underline{\underline{2 \text{ Words}}}$

Q.1

Consider the following program segment execute on Hypothetical processor.

[4 Marks]



Assume that program is stored in the memory address 1000 (Decimal) onwards. During the execution of I_6 what could be value present in the Program counter. Assume that word size is 32 bit & memory is Byte Addressable?

1 Word = 32 bit
= 4 Byte

3W x 4B
= 12 Byte

Instruction	Size (in words)
I_1	2
I_2	1
I_3	1
I_4	3w
I_5	1
I_6	2
I_7	1

Byte Addressable

1000 - 1007

1008 - 1011

1012 - 1015

1016 - 1027

1028 - 1031

1032 - 1039

1040 - 1043

1 Word = 4 Byte

2 Word = 2 x 4B

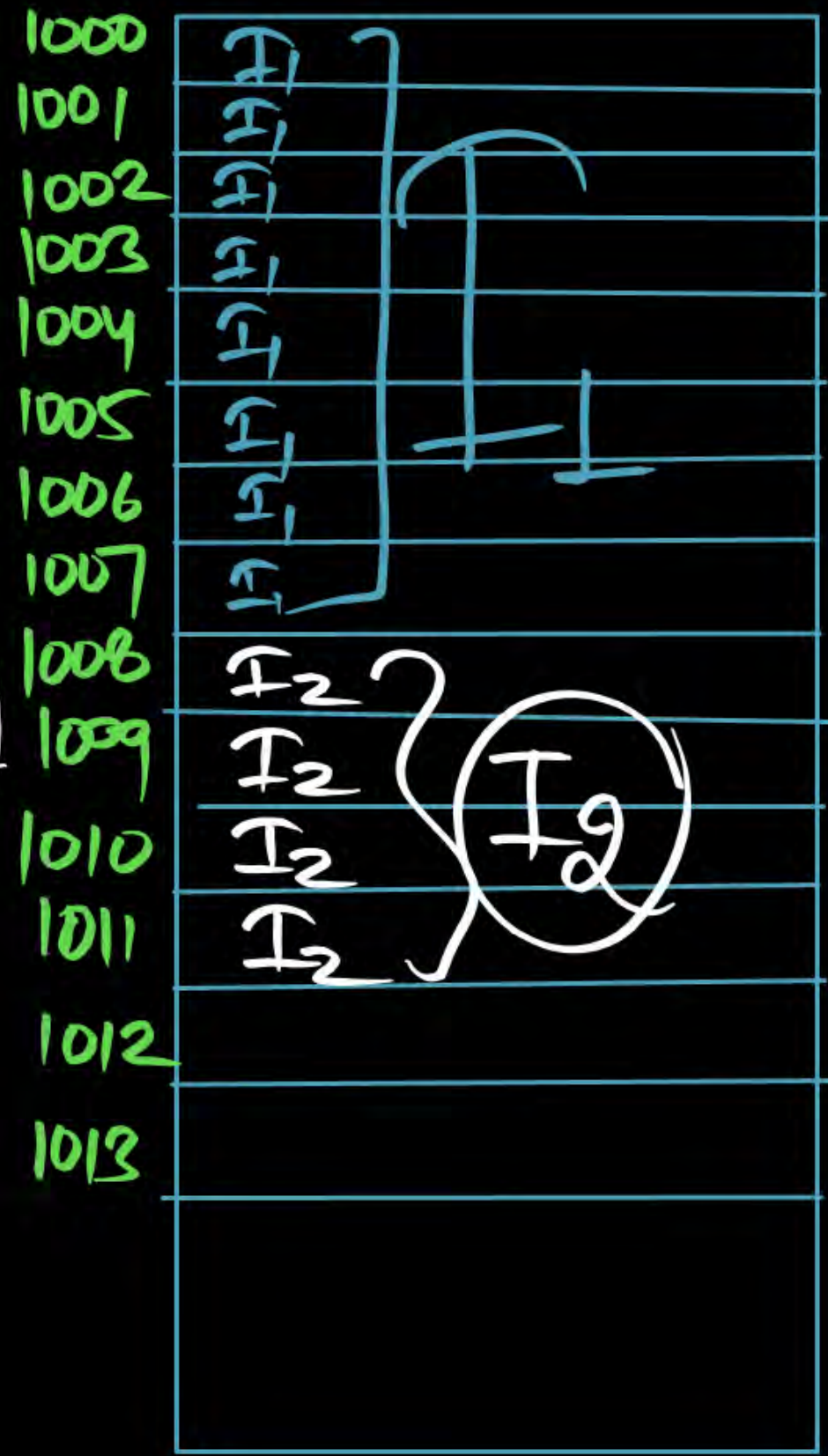
= 8 Byte

1 Word = 1 x 4B = 4 Byte

Ans 1

1040

Byte
Addressable



Q During the execution of I_6 What is PC Value?

Fetch
↓
Execute. PC Denote Next Instruction Starting Address.

Starting address of I_7 .

Q.2

Consider the following program segment execute on **[4 Marks]**
Hypothetical processor.

Assume that word size is 32 bit & memory is word addressable.
The program is stored in the memory at address 1000 (Decimal)
onwards. During the execution of I₅. What could be value present
in the program counter?

Instruction	Size (in words)
I ₁	2
I ₂	1
I ₃	1
I ₄	3
I ₅	1
I ₆	2
I ₇	1

Word Addressable.

1000 - 1001

1002

1003

1004, 5, 1006

1007

1008 - 1009

1010

Ans 2 1008

During Execution of I_5



If Interrupt occur During the execution of I_5 then what Return Address [value] Push into the Stack ?

Ans 1008

②

I₅

Interrupt occur. During execution of I₅

After I₅ Fetch PC Point (Denote) I₆ starting address

But Now I₅ encounter Interrupt

So PC Value (I₆ starting address)

as a Return address



Fetch

Execute

Interrupt

✓ [At the end of Fetch cycle]
After Fetch PC Denote Next Instrn Starting address.

$$\begin{aligned} 1 \text{ cell} &= 1 \text{ word} - \\ &= 32 \text{ bit} \end{aligned}$$

Word Addressable

1000	$\} I_1 (32 \text{ bit})$
1001	$\} I_1 (1 \text{ word})$
1002	$\} I_2$
1003	$\} I_3$
1004	
1005	

$I_1: 2 \text{ word}$

$I_2: 1 \text{ word}$

$I_3: 1 \text{ word}$

Q.



Consider the following Program Segment for a hypothetical CN.

Instruction	Meaning	Instruction size (in words)
I ₁ MOV r ₀ , 2000	$r_0 \leftarrow M[2000]$	3
I ₂ MOV r ₁ , 3000	$r_1 \leftarrow M[3000]$	3
I ₃ MUL r ₀ , r ₁	$r_0 \leftarrow r_0 * r_1$	1
I ₄ MOV 6000, r ₀	$M[6000] \leftarrow r_0$	3
I ₆ HALT	Machine Halt	1

51 cycle

Let the Clock Cycle required for various operation be as follows:

Instruction Fetch & Decode: 3 clock cycle per word

MUL with both operand & stored in register: 6 Clock Cycle.

Register to/from memory transfer: 4 clock cycle

The total number of clock cycle required to execute the program is___

Q.

Consider the following program segment for a hypothetical CPU Having three users registers R1, R2 and R3.

[GATE-2 Marks]

Instruction	Operation	Instruction size (in words)
MOV R1, 5000	$R1 \leftarrow \text{Memory}[5000]$	2
MOV R2, (R1)	$R2 \leftarrow \text{Memory}[(R1)]$	1
<u>ADD R2, R3</u>	$R2 \leftarrow R2 + R3$	1
MOV 6000, R2	$\text{Memory}[6000] \leftarrow R2$	2
HALT	Machine Halts	1

Consider that the memory is word addressable with size 32 bits and the program has been loaded starting from memory location 1000 (decimal). If an interrupt occurs during the ADD instruction, what will be the return address pushed on to the stack

(a) 1007

☒ (b) 1004

(c) 1005

(d) 1016

ADD Instⁿ (I₃) Fetch Cycle Done then PC Denote
the Next Instⁿ Starting address



PC Value $\xrightarrow{\text{Push into}}$ Stack.

$$1 \text{ Word} = 32 \text{ bit}$$

$$1 \text{ Word} = 4 \text{ Byte.}$$

$$I_1: 2 \text{ Word} \Rightarrow 2 \times 4 = 8 \text{ Byte}$$

Q.

Consider the following program segment for a hypothetical CPU Having three users registers R1, R2 and R3.

[GATE-
2 Marks]



Instruction	Operation	Instruction size (in words)
MOV R1, 5000	$R1 \leftarrow \text{Memory}[5000]$	2
MOV R2, (R1)	$R2 \leftarrow \text{Memory}[(R1)]$	1
ADD R2, R3	$R2 \leftarrow R2 + R3$	1
MOV 6000, R2	$\text{Memory}[6000] \leftarrow R2$	2
HALT	Machine Halts	1

Let the clock cycles required for various operations be as follows:

Register to/from memory transfer:

3. Clock cycles.

ADD with both operand in register

1. Clock cycle

Instruction fetch and decode:

2. Clock cycles per word.

The total number of clock cycle required to execute the program is

(a) 29 (b) 24 (c) 23 (d) 20

COMMON DATA QUESTION (5 - 7)

Consider the following program segment, Here R1, R2 and R3 are the general purpose register.

Instruction	Operation	Instruction size (no. of words)
MOV R1, (3000)	$R1 \leftarrow M[3000]$	2
LOOP;		
MOV R2, M[R3]	$R2 \leftarrow M[R3]$	1
ADD R2, R1	$R2 \leftarrow R1 + R2$	1
MOV (R3), R2	$M[R3] \leftarrow R2$	1
INC R3	$R3 \leftarrow R3 + 1$	1
DEC R1	$R1 \leftarrow R1 - 1$	1
BNZ LOOP	Branch on not zero	2
HALT		Stop

Assume that the content of memory location 3000 is 10 and the content of the Register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the Numbers are in decimal.

Q.5

Assume that the memory is word addressable. The number of memory references for accessing the data in executing the program completely is

[2 marks]

- (a) 10 (b) 11 (c) 20 (d) 21

Q.6

Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is

[2 marks]

- (a) 100 (b) 101 (c) 102 (d) 110

Q.7

Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on to the stack?

[2 marks]

(a) 1005

(b) 1020

(c) 1024

(d) 1040





**THANK
YOU!**

